
SOURCES AND USES OF FUNDS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date 12/31/2019
Delivery Date 12/31/2019

Sources:

Bond Proceeds:	
Par Amount	43,956,196.40
Premium	2,225,770.55
	46,181,966.95

Other Sources of Funds:	
Equity Contribution	3,980,294.55
	50,162,261.50

Uses:

Project Fund Deposits:	
Deposit to Bond Retirement Fund	373.21

Refunding Escrow Deposits:	
Cash Deposit	478.75
Open Market Purchases	49,787,914.43
	49,788,393.18

Delivery Date Expenses:	
Cost of Issuance	158,500.00
Underwriter's Discount	214,995.11
	373,495.11

50,162,261.50

SUMMARY OF REFUNDING RESULTS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date	12/31/2019
Delivery Date	12/31/2019
Arbitrage yield	3.506867%
Escrow yield	1.716992%
Value of Negative Arbitrage	3,045,298.26
Bond Par Amount	43,956,196.40
True Interest Cost	3.536303%
All-In TIC	3.558128%
Average Life	25.105
Weighted Average Maturity	24.240
Par amount of refunded bonds	43,960,000.00
Average coupon of refunded bonds	5.122776%
Average life of refunded bonds	26.135
Remaining weighted average maturity of refunded bonds	26.101
PV of prior debt to 12/31/2019 @ 3.506867%	55,937,371.03
Net PV Savings	5,775,482.74
Percentage savings of refunded bonds	13.138041%

BOND PRICING

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Bond Component	Maturity Date	Amount	Rate	Yield	Price	Principal per \$5,000 at Maturity	Offer Price per \$5,000 at Maturity	CAB Value at Maturity	Premium (-Discount)
Serial Bonds:									
	12/01/2020	5,000.00	4.000%	1.874%	101.92100000				96.05
	12/01/2021	5,000.00	4.000%	1.974%	103.79000000				189.50
	12/01/2030	470,000.00	2.748%	2.748%	100.00000000				
	12/01/2031	1,000,000.00	2.828%	2.828%	100.00000000				
	12/01/2032	1,035,000.00	2.908%	2.908%	100.00000000				
		<u>2,515,000.00</u>							<u>285.55</u>
Term Bond 2034:									
	12/01/2034	2,235,000.00	3.008%	3.008%	100.00000000				
Term Bond 2036:									
	12/01/2036	2,450,000.00	3.137%	3.137%	100.00000000				
Term Bond 2038:									
	12/01/2038	2,720,000.00	3.237%	3.237%	100.00000000				
Term Bond 2041:									
	12/01/2041	4,560,000.00	3.357%	3.357%	100.00000000				
Term Bond 2047:									
	12/01/2047	11,290,000.00	3.507%	3.507%	100.00000000				
Term Bond 2049:									
	12/01/2049	4,510,000.00	3.557%	3.557%	100.00000000				
Term Bond 2054:									
	12/01/2054	13,040,000.00	3.777%	3.777%	100.00000000				
Capital Appreciation Bonds:									
	12/01/2022	168,153.80	30.000%	2.249%	211.71272966	2,212.55	4,684.25	380,000	187,849.20
	12/01/2023	127,148.00	30.000%	2.351%	272.71966527	1,673.00	4,562.60	380,000	219,609.60
	12/01/2024	96,143.80	30.000%	2.431%	350.96636497	1,265.05	4,439.90	380,000	241,288.60
	12/01/2025	72,697.80	30.000%	2.642%	447.52495949	956.55	4,280.80	380,000	252,643.00
	12/01/2026	54,967.00	30.000%	2.742%	572.63048738	723.25	4,141.55	380,000	259,790.80

BOND PRICING

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Bond Component	Maturity Date	Amount	Rate	Yield	Price	Principal per \$5,000 at Maturity	Offer Price per \$5,000 at Maturity	CAB Value at Maturity	Premium (-Discount)
Capital Appreciation Bonds:									
	12/01/2027	41,564.40	30.000%	2.858%	730.28890108	546.90	3,993.95	380,000	261,975.80
	12/01/2028	31,426.00	30.000%	3.008%	926.56590085	413.50	3,831.35	380,000	259,756.60
	12/01/2029	23,765.20	30.000%	3.058%	1,183.40262232	312.70	3,700.50	380,000	257,472.80
	12/01/2030	20,330.40	30.000%	3.158%	1,502.32656514	236.40	3,551.50	430,000	285,098.60
		636,196.40						3,470,000	2,225,485.00
		43,956,196.40						3,470,000	2,225,770.55

BOND PRICING

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Takedown

3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
3.250
4.520
5.977
7.905
10.454
13.826
18.285
24.184
31.980
42.301

BOND PRICING

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date	12/31/2019	
Delivery Date	12/31/2019	
First Coupon	06/01/2020	
Par Amount	43,956,196.40	
Premium	2,225,770.55	
Production	46,181,966.95	105.063610%
Underwriter's Discount	-214,995.11	-0.489112%
Purchase Price	45,966,971.84	104.574498%
Accrued Interest		
Net Proceeds	45,966,971.84	

BOND SUMMARY STATISTICS

Brecksville-Broadview Heights City School District Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date	12/31/2019
Delivery Date	12/31/2019
Last Maturity	12/01/2054
Arbitrage Yield	3.506867%
True Interest Cost (TIC)	3.536303%
All-In TIC	3.558128%
Average Life (years)	25.105
Duration of Issue (years)	16.117
Par Amount	43,956,196.40
Bond Proceeds	46,181,966.95
Total Interest	39,022,174.70
Net Interest	37,011,399.26
Total Debt Service	85,812,174.70
Maximum Annual Debt Service	2,916,314.70
Average Annual Debt Service	2,457,432.42
Underwriter's Fees (per \$1000)	
Average Takedown	3.360846
Other Fee	1.530276
Total Underwriter's Discount	4.891122
Bid Price	104.574498

Bond Component	Par Value	Price	Average Coupon	Average Life	PV of 1 bp change
Capital Appreciation Bonds	636,196.40	449.811		5.250	1,903.80
Serial Bonds	2,515,000.00	100.011	2.850%	12.102	2,550.65
Term Bond 2034	2,235,000.00	100.000	3.008%	14.438	2,659.65
Term Bond 2036	2,450,000.00	100.000	3.137%	16.432	3,209.50
Term Bond 2038	2,720,000.00	100.000	3.237%	18.430	3,835.20
Term Bond 2041	4,560,000.00	100.000	3.357%	20.957	7,068.00
Term Bond 2047	11,290,000.00	100.000	3.507%	25.557	19,983.30
Term Bond 2049	4,510,000.00	100.000	3.557%	29.434	8,253.30
Term Bond 2054	13,040,000.00	100.000	3.777%	32.992	25,167.20
	43,956,196.40			25.105	74,630.60

BOND SUMMARY STATISTICS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

	TIC	All-In TIC	Arbitrage Yield
Par Value	43,956,196.40	43,956,196.40	43,956,196.40
+ Accrued Interest			
+ Premium (Discount)	2,225,770.55	2,225,770.55	2,225,770.55
- Underwriter's Discount	-214,995.11	-214,995.11	
- Cost of Issuance Expense		-158,500.00	
- Other Amounts			
Target Value	45,966,971.84	45,808,471.84	46,181,966.95
Target Date	12/31/2019	12/31/2019	12/31/2019
Yield	3.536303%	3.558128%	3.506867%

BOND MATURITY TABLE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Maturity Date	Serial Bonds	Term Bond 2034	Term Bond 2036	Term Bond 2038	Term Bond 2041	Term Bond 2047	Term Bond 2049	Term Bond 2054
12/01/2020	5,000							
12/01/2021	5,000							
12/01/2022								
12/01/2023								
12/01/2024								
12/01/2025								
12/01/2026								
12/01/2027								
12/01/2028								
12/01/2029								
12/01/2030	470,000							
12/01/2031	1,000,000							
12/01/2032	1,035,000							
12/01/2033		1,075,000						
12/01/2034		1,160,000						
12/01/2035			1,195,000					
12/01/2036			1,255,000					
12/01/2037				1,330,000				
12/01/2038				1,390,000				
12/01/2039					1,425,000			
12/01/2040					1,540,000			
12/01/2041					1,595,000			
12/01/2042						1,650,000		
12/01/2043						1,765,000		
12/01/2044						1,840,000		
12/01/2045						1,910,000		
12/01/2046						2,020,000		
12/01/2047						2,105,000		
12/01/2048							2,190,000	
12/01/2049							2,320,000	
12/01/2050								2,415,000

BOND MATURITY TABLE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Maturity Date	Serial Bonds	Term Bond 2034	Term Bond 2036	Term Bond 2038	Term Bond 2041	Term Bond 2047	Term Bond 2049	Term Bond 2054
12/01/2051								2,515,000
12/01/2052								2,610,000
12/01/2053								2,705,000
12/01/2054								2,795,000
	2,515,000	2,235,000	2,450,000	2,720,000	4,560,000	11,290,000	4,510,000	13,040,000

BOND MATURITY TABLE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Maturity Date	Capital Appreciation Bonds	Total
12/01/2020		5,000.00
12/01/2021		5,000.00
12/01/2022	168,153.80	168,153.80
12/01/2023	127,148.00	127,148.00
12/01/2024	96,143.80	96,143.80
12/01/2025	72,697.80	72,697.80
12/01/2026	54,967.00	54,967.00
12/01/2027	41,564.40	41,564.40
12/01/2028	31,426.00	31,426.00
12/01/2029	23,765.20	23,765.20
12/01/2030	20,330.40	490,330.40
12/01/2031		1,000,000.00
12/01/2032		1,035,000.00
12/01/2033		1,075,000.00
12/01/2034		1,160,000.00
12/01/2035		1,195,000.00
12/01/2036		1,255,000.00
12/01/2037		1,330,000.00
12/01/2038		1,390,000.00
12/01/2039		1,425,000.00
12/01/2040		1,540,000.00
12/01/2041		1,595,000.00
12/01/2042		1,650,000.00
12/01/2043		1,765,000.00
12/01/2044		1,840,000.00
12/01/2045		1,910,000.00
12/01/2046		2,020,000.00
12/01/2047		2,105,000.00
12/01/2048		2,190,000.00
12/01/2049		2,320,000.00
12/01/2050		2,415,000.00

BOND MATURITY TABLE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Maturity Date	Capital Appreciation Bonds	Total
12/01/2051		2,515,000.00
12/01/2052		2,610,000.00
12/01/2053		2,705,000.00
12/01/2054		2,795,000.00
	636,196.40	43,956,196.40

BOND DEBT SERVICE

**Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019**

Dated Date 12/31/2019
Delivery Date 12/31/2019

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
06/01/2020			631,593.61		631,593.61	
12/01/2020	5,000.00	4.000%	752,893.05		757,893.05	1,389,486.66
06/01/2021			752,793.05		752,793.05	
12/01/2021	5,000.00	4.000%	752,793.05		757,793.05	1,510,586.10
06/01/2022			752,693.05		752,693.05	
12/01/2022	168,153.80	30.000%	752,693.05	211,846.20	1,132,693.05	1,885,386.10
06/01/2023			752,693.05		752,693.05	
12/01/2023	127,148.00	30.000%	752,693.05	252,852.00	1,132,693.05	1,885,386.10
06/01/2024			752,693.05		752,693.05	
12/01/2024	96,143.80	30.000%	752,693.05	283,856.20	1,132,693.05	1,885,386.10
06/01/2025			752,693.05		752,693.05	
12/01/2025	72,697.80	30.000%	752,693.05	307,302.20	1,132,693.05	1,885,386.10
06/01/2026			752,693.05		752,693.05	
12/01/2026	54,967.00	30.000%	752,693.05	325,033.00	1,132,693.05	1,885,386.10
06/01/2027			752,693.05		752,693.05	
12/01/2027	41,564.40	30.000%	752,693.05	338,435.60	1,132,693.05	1,885,386.10
06/01/2028			752,693.05		752,693.05	
12/01/2028	31,426.00	30.000%	752,693.05	348,574.00	1,132,693.05	1,885,386.10
06/01/2029			752,693.05		752,693.05	
12/01/2029	23,765.20	30.000%	752,693.05	356,234.80	1,132,693.05	1,885,386.10
06/01/2030			752,693.05		752,693.05	
12/01/2030	490,330.40	** %	752,693.05	409,669.60	1,652,693.05	2,405,386.10
06/01/2031			746,235.25		746,235.25	
12/01/2031	1,000,000.00	2.828%	746,235.25		1,746,235.25	2,492,470.50
06/01/2032			732,095.25		732,095.25	
12/01/2032	1,035,000.00	2.908%	732,095.25		1,767,095.25	2,499,190.50
06/01/2033			717,046.35		717,046.35	
12/01/2033	1,075,000.00	3.008%	717,046.35		1,792,046.35	2,509,092.70
06/01/2034			700,878.35		700,878.35	
12/01/2034	1,160,000.00	3.008%	700,878.35		1,860,878.35	2,561,756.70
06/01/2035			683,431.95		683,431.95	
12/01/2035	1,195,000.00	3.137%	683,431.95		1,878,431.95	2,561,863.90
06/01/2036			664,688.38		664,688.38	
12/01/2036	1,255,000.00	3.137%	664,688.38		1,919,688.38	2,584,376.76
06/01/2037			645,003.70		645,003.70	
12/01/2037	1,330,000.00	3.237%	645,003.70		1,975,003.70	2,620,007.40
06/01/2038			623,477.65		623,477.65	
12/01/2038	1,390,000.00	3.237%	623,477.65		2,013,477.65	2,636,955.30
06/01/2039			600,980.50		600,980.50	
12/01/2039	1,425,000.00	3.357%	600,980.50		2,025,980.50	2,626,961.00
06/01/2040			577,061.88		577,061.88	
12/01/2040	1,540,000.00	3.357%	577,061.88		2,117,061.88	2,694,123.76
06/01/2041			551,212.98		551,212.98	
12/01/2041	1,595,000.00	3.357%	551,212.98		2,146,212.98	2,697,425.96
06/01/2042			524,440.90		524,440.90	
12/01/2042	1,650,000.00	3.507%	524,440.90		2,174,440.90	2,698,881.80
06/01/2043			495,508.15		495,508.15	
12/01/2043	1,765,000.00	3.507%	495,508.15		2,260,508.15	2,756,016.30
06/01/2044			464,558.88		464,558.88	
12/01/2044	1,840,000.00	3.507%	464,558.88		2,304,558.88	2,769,117.76
06/01/2045			432,294.48		432,294.48	

BOND DEBT SERVICE

Brecksville-Broadview Heights City School District Taxable School Facilities Improvement Refunding Bonds, Series 2019

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
12/01/2045	1,910,000.00	3.507%	432,294.48		2,342,294.48	2,774,588.96
06/01/2046			398,802.63		398,802.63	
12/01/2046	2,020,000.00	3.507%	398,802.63		2,418,802.63	2,817,605.26
06/01/2047			363,381.93		363,381.93	
12/01/2047	2,105,000.00	3.507%	363,381.93		2,468,381.93	2,831,763.86
06/01/2048			326,470.75		326,470.75	
12/01/2048	2,190,000.00	3.557%	326,470.75		2,516,470.75	2,842,941.50
06/01/2049			287,521.60		287,521.60	
12/01/2049	2,320,000.00	3.557%	287,521.60		2,607,521.60	2,895,043.20
06/01/2050			246,260.40		246,260.40	
12/01/2050	2,415,000.00	3.777%	246,260.40		2,661,260.40	2,907,520.80
06/01/2051			200,653.13		200,653.13	
12/01/2051	2,515,000.00	3.777%	200,653.13		2,715,653.13	2,916,306.26
06/01/2052			153,157.35		153,157.35	
12/01/2052	2,610,000.00	3.777%	153,157.35		2,763,157.35	2,916,314.70
06/01/2053			103,867.50		103,867.50	
12/01/2053	2,705,000.00	3.777%	103,867.50		2,808,867.50	2,912,735.00
06/01/2054			52,783.58		52,783.58	
12/01/2054	2,795,000.00	3.777%	52,783.58		2,847,783.58	2,900,567.16
	43,956,196.40		39,022,174.70	2,833,803.60	85,812,174.70	85,812,174.70

SAVINGS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Prior Debt Service	Refunding Debt Service	Savings	Annual Savings	Present Value to 12/31/2019 @ 3.5068670%
06/01/2020	1,120,750.00	631,593.61	489,156.39		482,075.32
12/01/2020	1,120,750.00	757,893.05	362,856.95	852,013.34	351,441.90
06/01/2021	1,120,750.00	752,793.05	367,956.95		350,240.23
12/01/2021	1,120,750.00	757,793.05	362,956.95	730,913.90	339,527.59
06/01/2022	1,120,750.00	752,693.05	368,056.95		338,365.36
12/01/2022	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-10,790.39
06/01/2023	1,120,750.00	752,693.05	368,056.95		326,804.29
12/01/2023	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-10,421.71
06/01/2024	1,120,750.00	752,693.05	368,056.95		315,638.24
12/01/2024	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-10,065.63
06/01/2025	1,120,750.00	752,693.05	368,056.95		304,853.69
12/01/2025	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-9,721.71
06/01/2026	1,120,750.00	752,693.05	368,056.95		294,437.63
12/01/2026	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-9,389.54
06/01/2027	1,120,750.00	752,693.05	368,056.95		284,377.46
12/01/2027	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-9,068.73
06/01/2028	1,120,750.00	752,693.05	368,056.95		274,661.02
12/01/2028	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-8,758.87
06/01/2029	1,120,750.00	752,693.05	368,056.95		265,276.56
12/01/2029	1,120,750.00	1,132,693.05	-11,943.05	356,113.90	-8,459.60
06/01/2030	1,120,750.00	752,693.05	368,056.95		256,212.75
12/01/2030	1,735,750.00	1,652,693.05	83,056.95	451,113.90	56,821.49
06/01/2031	1,105,375.00	746,235.25	359,139.75		241,463.25
12/01/2031	1,855,375.00	1,746,235.25	109,139.75	468,279.50	72,114.31
06/01/2032	1,086,625.00	732,095.25	354,529.75		230,219.50
12/01/2032	1,881,625.00	1,767,095.25	114,529.75	469,059.50	73,090.12
06/01/2033	1,066,750.00	717,046.35	349,703.65		219,326.67
12/01/2033	1,916,750.00	1,792,046.35	124,703.65	474,407.30	76,863.71
06/01/2034	1,045,500.00	700,878.35	344,621.65		208,754.42
12/01/2034	1,995,500.00	1,860,878.35	134,621.65	479,243.30	80,141.78
06/01/2035	1,021,750.00	683,431.95	338,318.05		197,933.89
12/01/2035	2,021,750.00	1,878,431.95	143,318.05	481,636.10	82,403.72
06/01/2036	996,750.00	664,688.38	332,061.62		187,635.72
12/01/2036	2,071,750.00	1,919,688.38	152,061.62	484,123.24	84,443.72
06/01/2037	969,875.00	645,003.70	324,871.30		177,300.53
12/01/2037	2,144,875.00	1,975,003.70	169,871.30	494,742.60	91,110.75
06/01/2038	940,500.00	623,477.65	317,022.35		167,105.38
12/01/2038	2,190,500.00	2,013,477.65	177,022.35	494,044.70	91,702.16
06/01/2039	909,250.00	600,980.50	308,269.50		156,939.75
12/01/2039	2,209,250.00	2,025,980.50	183,269.50	491,539.00	91,694.55
06/01/2040	876,750.00	577,061.88	299,688.12		147,358.02
12/01/2040	2,326,750.00	2,117,061.88	209,688.12	509,376.24	101,327.89
06/01/2041	840,500.00	551,212.98	289,287.02		137,383.65
12/01/2041	2,365,500.00	2,146,212.98	219,287.02	508,574.04	102,345.79
06/01/2042	802,375.00	524,440.90	277,934.10		127,482.27
12/01/2042	2,402,375.00	2,174,440.90	227,934.10	505,868.20	102,746.77
06/01/2043	762,375.00	495,508.15	266,866.85		118,223.67
12/01/2043	2,512,375.00	2,260,508.15	251,866.85	518,733.70	109,655.84
06/01/2044	718,625.00	464,558.88	254,066.12		108,707.23
12/01/2044	2,568,625.00	2,304,558.88	264,066.12	518,132.24	111,038.93
06/01/2045	672,375.00	432,294.48	240,080.52		99,213.43
12/01/2045	2,622,375.00	2,342,294.48	280,080.52	520,161.04	113,748.94
06/01/2046	623,625.00	398,802.63	224,822.37		89,733.56
12/01/2046	2,723,625.00	2,418,802.63	304,822.37	529,644.74	119,567.50

SAVINGS

Brecksville-Broadview Heights City School District Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Prior Debt Service	Refunding Debt Service	Savings	Annual Savings	Present Value to 12/31/2019 @ 3.5068670%
06/01/2047	571,125.00	363,381.93	207,743.07		80,083.64
12/01/2047	2,796,125.00	2,468,381.93	327,743.07	535,486.14	124,165.71
06/01/2048	515,500.00	326,470.75	189,029.25		70,379.81
12/01/2048	2,865,500.00	2,516,470.75	349,029.25	538,058.50	127,712.04
06/01/2049	456,750.00	287,521.60	169,228.40		60,854.71
12/01/2049	2,981,750.00	2,607,521.60	374,228.40	543,456.80	132,253.94
06/01/2050	390,468.75	246,260.40	144,208.35		50,085.64
12/01/2050	3,065,468.75	2,661,260.40	404,208.35	548,416.70	137,968.20
06/01/2051	320,250.00	200,653.13	119,596.87		40,118.48
12/01/2051	3,145,250.00	2,715,653.13	429,596.87	549,193.74	141,623.95
06/01/2052	246,093.75	153,157.35	92,936.40		30,110.11
12/01/2052	3,221,093.75	2,763,157.35	457,936.40	550,872.80	145,808.42
06/01/2053	168,000.00	103,867.50	64,132.50		20,068.11
12/01/2053	3,293,000.00	2,808,867.50	484,132.50	548,265.00	148,882.46
06/01/2054	85,968.75	52,783.58	33,185.17		10,029.38
12/01/2054	3,360,968.75	2,847,783.58	513,185.17	546,370.34	152,424.66
	103,002,812.50	85,812,174.70	17,190,637.80	17,190,637.80	9,755,404.08

Savings Summary

Dated Date	12/31/2019
Delivery Date	12/31/2019
PV of savings from cash flow	9,755,404.08
Less: Prior funds on hand	-3,980,294.55
Plus: Refunding funds on hand	373.21
Net PV Savings	5,775,482.74

SAVINGS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Prior Debt Service	Refunding Debt Service	Savings	Present Value to 12/31/2019 @ 3.5068670%
12/01/2020	2,241,500.00	1,389,486.66	852,013.34	833,517.22
12/01/2021	2,241,500.00	1,510,586.10	730,913.90	689,767.82
12/01/2022	2,241,500.00	1,885,386.10	356,113.90	327,574.97
12/01/2023	2,241,500.00	1,885,386.10	356,113.90	316,382.59
12/01/2024	2,241,500.00	1,885,386.10	356,113.90	305,572.61
12/01/2025	2,241,500.00	1,885,386.10	356,113.90	295,131.99
12/01/2026	2,241,500.00	1,885,386.10	356,113.90	285,048.09
12/01/2027	2,241,500.00	1,885,386.10	356,113.90	275,308.73
12/01/2028	2,241,500.00	1,885,386.10	356,113.90	265,902.15
12/01/2029	2,241,500.00	1,885,386.10	356,113.90	256,816.96
12/01/2030	2,856,500.00	2,405,386.10	451,113.90	313,034.24
12/01/2031	2,960,750.00	2,492,470.50	468,279.50	313,577.56
12/01/2032	2,968,250.00	2,499,190.50	469,059.50	303,309.62
12/01/2033	2,983,500.00	2,509,092.70	474,407.30	296,190.38
12/01/2034	3,041,000.00	2,561,756.70	479,243.30	288,896.20
12/01/2035	3,043,500.00	2,561,863.90	481,636.10	280,337.60
12/01/2036	3,068,500.00	2,584,376.76	484,123.24	272,079.45
12/01/2037	3,114,750.00	2,620,007.40	494,742.60	268,411.28
12/01/2038	3,131,000.00	2,636,955.30	494,044.70	258,807.54
12/01/2039	3,118,500.00	2,626,961.00	491,539.00	248,634.30
12/01/2040	3,203,500.00	2,694,123.76	509,376.24	248,685.92
12/01/2041	3,206,000.00	2,697,425.96	508,574.04	239,729.44
12/01/2042	3,204,750.00	2,698,881.80	505,868.20	230,229.05
12/01/2043	3,274,750.00	2,756,016.30	518,733.70	227,879.51
12/01/2044	3,287,250.00	2,769,117.76	518,132.24	219,746.17
12/01/2045	3,294,750.00	2,774,588.96	520,161.04	212,962.37
12/01/2046	3,347,250.00	2,817,605.26	529,644.74	209,301.06
12/01/2047	3,367,250.00	2,831,763.86	535,486.14	204,249.35
12/01/2048	3,381,000.00	2,842,941.50	538,058.50	198,091.85
12/01/2049	3,438,500.00	2,895,043.20	543,456.80	193,108.65
12/01/2050	3,455,937.50	2,907,520.80	548,416.70	188,053.84
12/01/2051	3,465,500.00	2,916,306.26	549,193.74	181,742.43
12/01/2052	3,467,187.50	2,916,314.70	550,872.80	175,918.53
12/01/2053	3,461,000.00	2,912,735.00	548,265.00	168,950.57
12/01/2054	3,446,937.50	2,900,567.16	546,370.34	162,454.05
	103,002,812.50	85,812,174.70	17,190,637.80	9,755,404.08

Savings Summary

Dated Date	12/31/2019
Delivery Date	12/31/2019
PV of savings from cash flow	9,755,404.08
Less: Prior funds on hand	-3,980,294.55
Plus: Refunding funds on hand	373.21
Net PV Savings	5,775,482.74

COST OF ISSUANCE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Cost of Issuance	\$/1000	Amount
Bond Counsel (Squire Patton Boggs)	1.80862	79,500.00
Municipal Advisor (Sudsina & Associates)	0.96687	42,500.00
Rating (Moody's)	0.65975	29,000.00
Paying Agent/Registrar (Huntington)	0.01137	500.00
Escrow Agent (Bank of New York Mellon)	0.04550	2,000.00
Verification Agent (Causey)	0.06825	3,000.00
Printer (DigitalMuni)	0.04550	2,000.00
	3.60586	158,500.00

UNDERWRITER'S DISCOUNT

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Underwriter's Discount	\$/1000	Amount
Average Takedown	3.36085	147,730.00
Continuing Disclosure Report	0.01365	600.00
Marketing Expenses	0.00967	425.25
Ipreo Bookrunning System	0.06180	2,716.49
Ipreo Wire Fees	0.00205	90.00
Ipreo Order Monitor/Dealer EOE	0.03000	1,318.69
DTC Setup	0.01820	800.00
OMAC	0.11375	5,000.00
CUSIP Numbers	0.01586	697.00
Day Loan	0.02614	1,149.18
Management Fee	1.23915	54,468.50
	4.89112	214,995.11

AVERAGE TAKEDOWN

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date 12/31/2019
Delivery Date 12/31/2019

Bond Component	Maturity Date	Par Amount	Takedown \$/Bond	Takedown Amount
Serial Bonds:				
	12/01/2020	5,000.00	3.2500	16.25
	12/01/2021	5,000.00	3.2500	16.25
	12/01/2030	470,000.00	3.2500	1,527.50
	12/01/2031	1,000,000.00	3.2500	3,250.00
	12/01/2032	1,035,000.00	3.2500	3,363.75
		<u>2,515,000.00</u>	<u>3.2500</u>	<u>8,173.75</u>
Term Bond 2034:				
	12/01/2034	2,235,000.00	3.2500	7,263.75
Term Bond 2036:				
	12/01/2036	2,450,000.00	3.2500	7,962.50
Term Bond 2038:				
	12/01/2038	2,720,000.00	3.2500	8,840.00
Term Bond 2041:				
	12/01/2041	4,560,000.00	3.2500	14,820.00
Term Bond 2047:				
	12/01/2047	11,290,000.00	3.2500	36,692.50
Term Bond 2049:				
	12/01/2049	4,510,000.00	3.2500	14,657.50
Term Bond 2054:				
	12/01/2054	13,040,000.00	3.2500	42,380.00
Capital Appreciation Bonds:				
	12/01/2022	168,153.80	2.0000	760.00
	12/01/2023	127,148.00	2.0000	760.00
	12/01/2024	96,143.80	2.0000	760.00
	12/01/2025	72,697.80	2.0000	760.00
	12/01/2026	54,967.00	2.0000	760.00
	12/01/2027	41,564.40	2.0000	760.00
	12/01/2028	31,426.00	2.0000	760.00
	12/01/2029	23,765.20	2.0000	760.00
	12/01/2030	20,330.40	2.0000	860.00
		<u>636,196.40</u>	<u>2.0000</u>	<u>6,940.00</u>
		43,956,196.40	3.2319	147,730.00

BOND ACCRETED VALUE TABLE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Capital Appreciation Bonds 12/01/2022 30%	Capital Appreciation Bonds 12/01/2023 30%	Capital Appreciation Bonds 12/01/2024 30%	Capital Appreciation Bonds 12/01/2025 30%	Capital Appreciation Bonds 12/01/2026 30%	Capital Appreciation Bonds 12/01/2027 30%	Capital Appreciation Bonds 12/01/2028 30%	Capital Appreciation Bonds 12/01/2029 30%	Capital Appreciation Bonds 12/01/2030 30%
12/31/2019	2,212.55	1,673.00	1,265.05	956.55	723.25	546.90	413.50	312.70	236.40
06/01/2020	2,485.85	1,879.65	1,421.30	1,074.70	812.60	614.45	464.60	351.30	265.65
12/01/2020	2,858.75	2,161.60	1,634.50	1,235.90	934.50	706.60	534.30	404.00	305.50
06/01/2021	3,287.55	2,485.85	1,879.65	1,421.30	1,074.70	812.60	614.45	464.60	351.30
12/01/2021	3,780.70	2,858.75	2,161.60	1,634.50	1,235.90	934.50	706.60	534.30	404.00
06/01/2022	4,347.80	3,287.55	2,485.85	1,879.65	1,421.30	1,074.70	812.60	614.45	464.60
12/01/2022	5,000.00	3,780.70	2,858.75	2,161.60	1,634.50	1,235.90	934.50	706.60	534.30
06/01/2023		4,347.80	3,287.55	2,485.85	1,879.65	1,421.30	1,074.70	812.60	614.45
12/01/2023		5,000.00	3,780.70	2,858.75	2,161.60	1,634.50	1,235.90	934.50	706.60
06/01/2024			4,347.80	3,287.55	2,485.85	1,879.65	1,421.30	1,074.70	812.60
12/01/2024			5,000.00	3,780.70	2,858.75	2,161.60	1,634.50	1,235.90	934.50
06/01/2025				4,347.80	3,287.55	2,485.85	1,879.65	1,421.30	1,074.70
12/01/2025				5,000.00	3,780.70	2,858.75	2,161.60	1,634.50	1,235.90
06/01/2026					4,347.80	3,287.55	2,485.85	1,879.65	1,421.30
12/01/2026					5,000.00	3,780.70	2,858.75	2,161.60	1,634.50
06/01/2027						4,347.80	3,287.55	2,485.85	1,879.65
12/01/2027						5,000.00	3,780.70	2,858.75	2,161.60
06/01/2028							4,347.80	3,287.55	2,485.85
12/01/2028							5,000.00	3,780.70	2,858.75
06/01/2029								4,347.80	3,287.55
12/01/2029								5,000.00	3,780.70
06/01/2030									4,347.80
12/01/2030									5,000.00

PRIOR BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Period Ending	Principal	Coupon	Interest	Debt Service	Annual Debt Service
06/01/2020			1,120,750.00	1,120,750.00	
12/01/2020			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2021			1,120,750.00	1,120,750.00	
12/01/2021			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2022			1,120,750.00	1,120,750.00	
12/01/2022			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2023			1,120,750.00	1,120,750.00	
12/01/2023			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2024			1,120,750.00	1,120,750.00	
12/01/2024			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2025			1,120,750.00	1,120,750.00	
12/01/2025			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2026			1,120,750.00	1,120,750.00	
12/01/2026			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2027			1,120,750.00	1,120,750.00	
12/01/2027			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2028			1,120,750.00	1,120,750.00	
12/01/2028			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2029			1,120,750.00	1,120,750.00	
12/01/2029			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2030			1,120,750.00	1,120,750.00	
12/01/2030	615,000	5.000%	1,120,750.00	1,735,750.00	2,856,500.00
06/01/2031			1,105,375.00	1,105,375.00	
12/01/2031	750,000	5.000%	1,105,375.00	1,855,375.00	2,960,750.00
06/01/2032			1,086,625.00	1,086,625.00	
12/01/2032	795,000	5.000%	1,086,625.00	1,881,625.00	2,968,250.00
06/01/2033			1,066,750.00	1,066,750.00	
12/01/2033	850,000	5.000%	1,066,750.00	1,916,750.00	2,983,500.00
06/01/2034			1,045,500.00	1,045,500.00	
12/01/2034	950,000	5.000%	1,045,500.00	1,995,500.00	3,041,000.00
06/01/2035			1,021,750.00	1,021,750.00	
12/01/2035	1,000,000	5.000%	1,021,750.00	2,021,750.00	3,043,500.00
06/01/2036			996,750.00	996,750.00	
12/01/2036	1,075,000	5.000%	996,750.00	2,071,750.00	3,068,500.00
06/01/2037			969,875.00	969,875.00	
12/01/2037	1,175,000	5.000%	969,875.00	2,144,875.00	3,114,750.00
06/01/2038			940,500.00	940,500.00	
12/01/2038	1,250,000	5.000%	940,500.00	2,190,500.00	3,131,000.00
06/01/2039			909,250.00	909,250.00	
12/01/2039	1,300,000	5.000%	909,250.00	2,209,250.00	3,118,500.00
06/01/2040			876,750.00	876,750.00	
12/01/2040	1,450,000	5.000%	876,750.00	2,326,750.00	3,203,500.00
06/01/2041			840,500.00	840,500.00	
12/01/2041	1,525,000	5.000%	840,500.00	2,365,500.00	3,206,000.00
06/01/2042			802,375.00	802,375.00	
12/01/2042	1,600,000	5.000%	802,375.00	2,402,375.00	3,204,750.00
06/01/2043			762,375.00	762,375.00	
12/01/2043	1,750,000	5.000%	762,375.00	2,512,375.00	3,274,750.00
06/01/2044			718,625.00	718,625.00	
12/01/2044	1,850,000	5.000%	718,625.00	2,568,625.00	3,287,250.00
06/01/2045			672,375.00	672,375.00	
12/01/2045	1,950,000	5.000%	672,375.00	2,622,375.00	3,294,750.00
06/01/2046			623,625.00	623,625.00	
12/01/2046	2,100,000	5.000%	623,625.00	2,723,625.00	3,347,250.00
06/01/2047			571,125.00	571,125.00	

PRIOR BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Period Ending	Principal	Coupon	Interest	Debt Service	Annual Debt Service
12/01/2047	2,225,000	5.000%	571,125.00	2,796,125.00	3,367,250.00
06/01/2048			515,500.00	515,500.00	
12/01/2048	2,350,000	5.000%	515,500.00	2,865,500.00	3,381,000.00
06/01/2049			456,750.00	456,750.00	
12/01/2049	2,525,000	5.250%	456,750.00	2,981,750.00	3,438,500.00
06/01/2050			390,468.75	390,468.75	
12/01/2050	2,675,000	5.250%	390,468.75	3,065,468.75	3,455,937.50
06/01/2051			320,250.00	320,250.00	
12/01/2051	2,825,000	5.250%	320,250.00	3,145,250.00	3,465,500.00
06/01/2052			246,093.75	246,093.75	
12/01/2052	2,975,000	5.250%	246,093.75	3,221,093.75	3,467,187.50
06/01/2053			168,000.00	168,000.00	
12/01/2053	3,125,000	5.250%	168,000.00	3,293,000.00	3,461,000.00
06/01/2054			85,968.75	85,968.75	
12/01/2054	3,275,000	5.250%	85,968.75	3,360,968.75	3,446,937.50
	43,960,000		59,042,812.50	103,002,812.50	103,002,812.50

SUMMARY OF BONDS REFUNDED

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Bond	Maturity Date	Interest Rate	Par Amount	Call Date	Call Price
School Facilities Improvement Bonds, Series 2018, 2018:					
SERIAL	12/01/2030	5.000%	615,000.00	12/01/2023	100.000
	12/01/2031	5.000%	750,000.00	12/01/2023	100.000
	12/01/2032	5.000%	795,000.00	12/01/2023	100.000
	12/01/2033	5.000%	850,000.00	12/01/2023	100.000
	12/01/2034	5.000%	950,000.00	12/01/2023	100.000
	12/01/2035	5.000%	1,000,000.00	12/01/2023	100.000
	12/01/2036	5.000%	1,075,000.00	12/01/2023	100.000
	12/01/2037	5.000%	1,175,000.00	12/01/2023	100.000
	12/01/2038	5.000%	1,250,000.00	12/01/2023	100.000
	12/01/2045	5.000%	11,425,000.00	12/01/2023	100.000
TERM45	12/01/2045	5.000%	11,425,000.00	12/01/2023	100.000
TERM48	12/01/2048	5.000%	6,675,000.00	12/01/2023	100.000
TERM54	12/01/2054	5.250%	17,400,000.00	12/01/2023	100.000
			43,960,000.00		

SUMMARY OF UNREFUNDED BONDS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Bond	Maturity Date	Interest Rate	Par Amount	Value on Dec 31, 2019
School Facilities Improvement Bonds, Series 2018, 2018:				
SERIAL	12/01/2020	2.000%	5,000.00	5,000.00
	12/01/2021	2.000%	5,000.00	5,000.00
CAB	12/01/2022	29.924%	113,985.90	164,043.20
	12/01/2023	29.924%	94,405.50	135,865.35
	12/01/2024	29.924%	78,484.65	112,949.90
	12/01/2025	29.924%	74,732.00	107,548.00
	12/01/2026	29.925%	57,048.05	82,100.15
	12/01/2027	29.924%	43,930.00	63,221.25
	12/01/2028	29.926%	36,703.00	52,819.30
	12/01/2029	29.924%	28,868.40	41,540.40
	12/01/2030	29.926%	1,819.95	2,619.10
			539,977.45	772,706.65

UNREFUNDED BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
06/01/2020			100		100	
12/01/2020	5,000.00	2.00000000%	100		5,100	5,200
06/01/2021			50		50	
12/01/2021	5,000.00	2.00000000%	50		5,050	5,100
12/01/2022	113,985.90	29.92419306%		256,014.10	370,000	370,000
12/01/2023	94,405.50	29.92408076%		310,594.50	405,000	405,000
12/01/2024	78,484.65	29.92444800%		366,515.35	445,000	445,000
12/01/2025	74,732.00	29.92429490%		485,268.00	560,000	560,000
12/01/2026	57,048.05	29.92482568%		507,951.95	565,000	565,000
12/01/2027	43,930.00	29.92445992%		531,070.00	575,000	575,000
12/01/2028	36,703.00	29.92588836%		598,297.00	635,000	635,000
12/01/2029	28,868.40	29.92431514%		631,131.60	660,000	660,000
12/01/2030	1,819.95	29.92587886%		53,180.05	55,000	55,000
	539,977.45		300	3,740,022.55	4,280,300	4,280,300

AGGREGATE DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Period Ending	Taxable School Facilities Improvement Refunding Bonds, Series 2019	Unrefunded Bonds	Aggregate Debt Service
12/01/2020	1,389,486.66	5,200	1,394,686.66
12/01/2021	1,510,586.10	5,100	1,515,686.10
12/01/2022	1,885,386.10	370,000	2,255,386.10
12/01/2023	1,885,386.10	405,000	2,290,386.10
12/01/2024	1,885,386.10	445,000	2,330,386.10
12/01/2025	1,885,386.10	560,000	2,445,386.10
12/01/2026	1,885,386.10	565,000	2,450,386.10
12/01/2027	1,885,386.10	575,000	2,460,386.10
12/01/2028	1,885,386.10	635,000	2,520,386.10
12/01/2029	1,885,386.10	660,000	2,545,386.10
12/01/2030	2,405,386.10	55,000	2,460,386.10
12/01/2031	2,492,470.50		2,492,470.50
12/01/2032	2,499,190.50		2,499,190.50
12/01/2033	2,509,092.70		2,509,092.70
12/01/2034	2,561,756.70		2,561,756.70
12/01/2035	2,561,863.90		2,561,863.90
12/01/2036	2,584,376.76		2,584,376.76
12/01/2037	2,620,007.40		2,620,007.40
12/01/2038	2,636,955.30		2,636,955.30
12/01/2039	2,626,961.00		2,626,961.00
12/01/2040	2,694,123.76		2,694,123.76
12/01/2041	2,697,425.96		2,697,425.96
12/01/2042	2,698,881.80		2,698,881.80
12/01/2043	2,756,016.30		2,756,016.30
12/01/2044	2,769,117.76		2,769,117.76
12/01/2045	2,774,588.96		2,774,588.96
12/01/2046	2,817,605.26		2,817,605.26
12/01/2047	2,831,763.86		2,831,763.86
12/01/2048	2,842,941.50		2,842,941.50
12/01/2049	2,895,043.20		2,895,043.20
12/01/2050	2,907,520.80		2,907,520.80
12/01/2051	2,916,306.26		2,916,306.26
12/01/2052	2,916,314.70		2,916,314.70
12/01/2053	2,912,735.00		2,912,735.00
12/01/2054	2,900,567.16		2,900,567.16
	85,812,174.70	4,280,300	90,092,474.70

ESCROW REQUIREMENTS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date 12/31/2019
Delivery Date 12/31/2019

Period Ending	Interest	Principal Redeemed	Total
06/01/2020	1,120,750.00		1,120,750.00
12/01/2020	1,120,750.00		1,120,750.00
06/01/2021	1,120,750.00		1,120,750.00
12/01/2021	1,120,750.00		1,120,750.00
06/01/2022	1,120,750.00		1,120,750.00
12/01/2022	1,120,750.00		1,120,750.00
06/01/2023	1,120,750.00		1,120,750.00
12/01/2023	1,120,750.00	43,960,000.00	45,080,750.00
	8,966,000.00	43,960,000.00	52,926,000.00

ESCROW COST

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Type of Security	Maturity Date	Par Amount	Rate	Yield	Price	Cost	Accrued Interest	Total Cost
TNote	05/31/2020	1,042,000	1.500%	1.573739%	99.968750	1,041,674.38	1,323.85	1,042,998.23
TNote	11/30/2020	1,050,000	2.750%	1.626739%	101.015625	1,060,664.06	2,445.70	1,063,109.76
TNote	05/31/2021	1,064,000	2.000%	1.640741%	100.500000	1,069,320.00	1,802.40	1,071,122.40
TNote	11/30/2021	1,076,000	1.875%	1.633448%	100.453125	1,080,875.63	1,708.81	1,082,584.44
TNote	05/31/2022	1,085,000	1.750%	1.617341%	100.312500	1,088,390.63	1,608.23	1,089,998.86
TNote	11/30/2022	1,094,000	2.000%	1.641590%	101.015625	1,105,110.94	1,853.22	1,106,964.16
TNote	05/31/2023	1,106,000	2.750%	1.672233%	103.562500	1,145,401.25	2,576.13	1,147,977.38
TSTRIP-I	11/15/2023	45,080,000		1.721982%	93.574000	42,183,159.20		42,183,159.20
		52,597,000				49,774,596.09	13,318.34	49,787,914.43

Purchase Date	Cost of Securities	Cash Deposit	Total Escrow Cost
12/31/2019	49,787,914.43	478.75	49,788,393.18
	49,787,914.43	478.75	49,788,393.18

ESCROW DESCRIPTIONS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Type of Security	CUSIP or ID	Maturity Date	First Int Pmt Date	Par Amount	Rate	Max Rate	Yield	Price	Dated Date	Interest Class	Interest Frequency	Interest Day Basis
Dec 31, 2019:												
TNote	912828XE5	05/31/2020	05/31/2020	1,042,000	1.500%	1.580%	1.574%	99.968750	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	9128285Q9	11/30/2020	05/31/2020	1,050,000	2.750%	1.530%	1.627%	101.015625	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	912828WN6	05/31/2021	05/31/2020	1,064,000	2.000%	1.570%	1.641%	100.500000	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	912828G53	11/30/2021	05/31/2020	1,076,000	1.875%	1.600%	1.633%	100.453125	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	912828XR6	05/31/2022	05/31/2020	1,085,000	1.750%	1.610%	1.617%	100.312500	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	912828M80	11/30/2022	05/31/2020	1,094,000	2.000%	1.620%	1.642%	101.015625	11/30/2019	Periodic	Semiannual	ACT/ACT
TNote	9128284S6	05/31/2023	05/31/2020	1,106,000	2.750%	1.640%	1.672%	103.562500	11/30/2019	Periodic	Semiannual	ACT/ACT
TSTRIP-I	912833LP3	11/15/2023		45,080,000		1.650%	1.722%	93.574000		Zero Coupon	Semiannual	ACT/ACT
				52,597,000								

ESCROW DESCRIPTIONS DETAIL

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

	Type of Security	CUSIP or ID	Maturity Date	First Int Pmt Date	Par Amount	Rate	Max Rate	Yield	Price	Dated Date	Interest Class	Interest Frequency
EQUITY, Dec 31, 2019:												
	TNote	912828XE5	05/31/2020	05/31/2020	1,042,000.00	1.500%	1.580%	1.574%	99.968750	11/30/2019	Periodic	Semiannual
	TNote	9128285Q9	11/30/2020	05/31/2020	1,050,000.00	2.750%	1.530%	1.627%	101.015625	11/30/2019	Periodic	Semiannual
	TNote	912828WN6	05/31/2021	05/31/2020	1,064,000.00	2.000%	1.570%	1.641%	100.500000	11/30/2019	Periodic	Semiannual
	TNote	912828G53	11/30/2021	05/31/2020	797,703.96	1.875%	1.600%	1.633%	100.453125	11/30/2019	Periodic	Semiannual
					3,953,703.96							
ESC, Dec 31, 2019:												
	TNote	912828G53	11/30/2021	05/31/2020	278,296.04	1.875%	1.600%	1.633%	100.453125	11/30/2019	Periodic	Semiannual
	TNote	912828XR6	05/31/2022	05/31/2020	1,085,000.00	1.750%	1.610%	1.617%	100.312500	11/30/2019	Periodic	Semiannual
	TNote	912828M80	11/30/2022	05/31/2020	1,094,000.00	2.000%	1.620%	1.642%	101.015625	11/30/2019	Periodic	Semiannual
	TNote	9128284S6	05/31/2023	05/31/2020	1,106,000.00	2.750%	1.640%	1.672%	103.562500	11/30/2019	Periodic	Semiannual
	TSTRIP-I	912833LP3	11/15/2023		45,080,000.00		1.650%	1.722%	93.574000		Zero Coupon	Semiannual
					48,643,296.04							
					52,597,000.00							

ESCROW DESCRIPTIONS DETAIL

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

	Maturity Date	Interest Day Basis
EQUITY, Dec 31, 2019:		
	05/31/2020	ACT/ACT
	11/30/2020	ACT/ACT
	05/31/2021	ACT/ACT
	11/30/2021	ACT/ACT

ESC, Dec 31, 2019:		
	11/30/2021	ACT/ACT
	05/31/2022	ACT/ACT
	11/30/2022	ACT/ACT
	05/31/2023	ACT/ACT
	11/15/2023	ACT/ACT

ESCROW CASH FLOW

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Principal	Interest	Net Escrow Receipts
05/31/2020	1,042,000.00	78,621.25	1,120,621.25
11/30/2020	1,050,000.00	70,806.25	1,120,806.25
05/31/2021	1,064,000.00	56,368.75	1,120,368.75
11/30/2021	1,076,000.00	45,728.75	1,121,728.75
05/31/2022	1,085,000.00	35,641.25	1,120,641.25
11/30/2022	1,094,000.00	26,147.50	1,120,147.50
05/31/2023	1,106,000.00	15,207.50	1,121,207.50
11/15/2023	45,080,000.00		45,080,000.00
	52,597,000.00	328,521.25	52,925,521.25

Escrow Cost Summary

Purchase date	12/31/2019
Purchase cost of securities	49,787,914.43

ESCROW STATISTICS

Brecksville-Broadview Heights City School District Taxable School Facilities Improvement Refunding Bonds, Series 2019

Escrow	Total Escrow Cost	Modified Duration (years)	PV of 1 bp change	Yield to Receipt Date	Yield to Disbursement Date	Perfect Escrow Cost	Value of Negative Arbitrage	Cost of Dead Time
EQUITY	3,980,294.55	1.102	438.39	1.626690%	1.622615%	3,898,896.82	81,014.32	383.41
ESC	45,808,098.64	3.755	17,198.42	1.716992%	1.698441%	42,782,594.99	2,964,283.94	61,219.71
	49,788,393.19		17,636.81			46,681,491.81	3,045,298.26	61,603.12

Delivery date 12/31/2019
Arbitrage yield 3.506867%

ESCROW SUFFICIENCY

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Date	Escrow Requirement	Net Escrow Receipts	Excess Receipts	Excess Balance
12/31/2019		478.75	478.75	478.75
05/31/2020		1,120,621.25	1,120,621.25	1,121,100.00
06/01/2020	1,120,750.00		-1,120,750.00	350.00
11/30/2020		1,120,806.25	1,120,806.25	1,121,156.25
12/01/2020	1,120,750.00		-1,120,750.00	406.25
05/31/2021		1,120,368.75	1,120,368.75	1,120,775.00
06/01/2021	1,120,750.00		-1,120,750.00	25.00
11/30/2021		1,121,728.75	1,121,728.75	1,121,753.75
12/01/2021	1,120,750.00		-1,120,750.00	1,003.75
05/31/2022		1,120,641.25	1,120,641.25	1,121,645.00
06/01/2022	1,120,750.00		-1,120,750.00	895.00
11/30/2022		1,120,147.50	1,120,147.50	1,121,042.50
12/01/2022	1,120,750.00		-1,120,750.00	292.50
05/31/2023		1,121,207.50	1,121,207.50	1,121,500.00
06/01/2023	1,120,750.00		-1,120,750.00	750.00
11/15/2023		45,080,000.00	45,080,000.00	45,080,750.00
12/01/2023	45,080,750.00		-45,080,750.00	
	52,926,000.00	52,926,000.00	0.00	

FORM 8038 STATISTICS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Dated Date 12/31/2019
Delivery Date 12/31/2019

Bond Component	Date	Principal	Coupon	Price	Issue Price	Redemption at Maturity
Serial Bonds:						
	12/01/2020	5,000.00	4.000%	101.921	5,096.05	5,000.00
	12/01/2021	5,000.00	4.000%	103.790	5,189.50	5,000.00
	12/01/2030	470,000.00	2.748%	100.000	470,000.00	470,000.00
	12/01/2031	1,000,000.00	2.828%	100.000	1,000,000.00	1,000,000.00
	12/01/2032	1,035,000.00	2.908%	100.000	1,035,000.00	1,035,000.00
Term Bond 2034:						
	12/01/2033	1,075,000.00	3.008%	100.000	1,075,000.00	1,075,000.00
	12/01/2034	1,160,000.00	3.008%	100.000	1,160,000.00	1,160,000.00
Term Bond 2036:						
	12/01/2035	1,195,000.00	3.137%	100.000	1,195,000.00	1,195,000.00
	12/01/2036	1,255,000.00	3.137%	100.000	1,255,000.00	1,255,000.00
Term Bond 2038:						
	12/01/2037	1,330,000.00	3.237%	100.000	1,330,000.00	1,330,000.00
	12/01/2038	1,390,000.00	3.237%	100.000	1,390,000.00	1,390,000.00
Term Bond 2041:						
	12/01/2039	1,425,000.00	3.357%	100.000	1,425,000.00	1,425,000.00
	12/01/2040	1,540,000.00	3.357%	100.000	1,540,000.00	1,540,000.00
	12/01/2041	1,595,000.00	3.357%	100.000	1,595,000.00	1,595,000.00
Term Bond 2047:						
	12/01/2042	1,650,000.00	3.507%	100.000	1,650,000.00	1,650,000.00
	12/01/2043	1,765,000.00	3.507%	100.000	1,765,000.00	1,765,000.00
	12/01/2044	1,840,000.00	3.507%	100.000	1,840,000.00	1,840,000.00
	12/01/2045	1,910,000.00	3.507%	100.000	1,910,000.00	1,910,000.00
	12/01/2046	2,020,000.00	3.507%	100.000	2,020,000.00	2,020,000.00
	12/01/2047	2,105,000.00	3.507%	100.000	2,105,000.00	2,105,000.00
Term Bond 2049:						
	12/01/2048	2,190,000.00	3.557%	100.000	2,190,000.00	2,190,000.00
	12/01/2049	2,320,000.00	3.557%	100.000	2,320,000.00	2,320,000.00
Term Bond 2054:						
	12/01/2050	2,415,000.00	3.777%	100.000	2,415,000.00	2,415,000.00
	12/01/2051	2,515,000.00	3.777%	100.000	2,515,000.00	2,515,000.00
	12/01/2052	2,610,000.00	3.777%	100.000	2,610,000.00	2,610,000.00
	12/01/2053	2,705,000.00	3.777%	100.000	2,705,000.00	2,705,000.00
	12/01/2054	2,795,000.00	3.777%	100.000	2,795,000.00	2,795,000.00
Capital Appreciation Bonds:						
	12/01/2022	168,153.80	30.000%	211.713	356,003.00	380,000.00
	12/01/2023	127,148.00	30.000%	272.720	346,757.60	380,000.00
	12/01/2024	96,143.80	30.000%	350.966	337,432.40	380,000.00
	12/01/2025	72,697.80	30.000%	447.525	325,340.80	380,000.00
	12/01/2026	54,967.00	30.000%	572.630	314,757.80	380,000.00

FORM 8038 STATISTICS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Bond Component	Date	Principal	Coupon	Price	Issue Price	Redemption at Maturity
Capital Appreciation Bonds:						
	12/01/2027	41,564.40	30.000%	730.289	303,540.20	380,000.00
	12/01/2028	31,426.00	30.000%	926.566	291,182.60	380,000.00
	12/01/2029	23,765.20	30.000%	1,183.403	281,238.00	380,000.00
	12/01/2030	20,330.40	30.000%	1,502.327	305,429.00	430,000.00
		43,956,196.40			46,181,966.95	46,790,000.00

	Maturity Date	Interest Rate	Issue Price	Stated Redemption at Maturity	Weighted Average Maturity	Yield
Final Maturity	12/01/2054	3.777%	2,795,000.00	2,795,000.00		
Entire Issue			46,181,966.95	46,790,000.00	24.2401	3.5069%

Proceeds used for accrued interest	0.00
Proceeds used for bond issuance costs (including underwriters' discount)	373,495.11
Proceeds used for credit enhancement	0.00
Proceeds allocated to reasonably required reserve or replacement fund	0.00
Proceeds used to refund prior tax-exempt bonds	45,808,098.64
Proceeds used to refund prior taxable bonds	0.00
Remaining WAM of prior tax-exempt bonds (years)	26.1005
Remaining WAM of prior taxable bonds (years)	0.0000
Last call date of refunded tax-exempt bonds	12/01/2023

2011 Form 8038 Statistics

Proceeds used to currently refund prior issues	0.00
Proceeds used to advance refund prior issues	45,808,098.64
Remaining weighted average maturity of the bonds to be currently refunded	0.0000
Remaining weighted average maturity of the bonds to be advance refunded	26.1005

FORM 8038 STATISTICS

Brecksville-Broadview Heights City School District
Taxable School Facilities Improvement Refunding Bonds, Series 2019

Refunded Bonds

Bond Component	Date	Principal	Coupon	Price	Issue Price
School Facilities Improvement Bonds, Series 2018:					
SERIAL	12/01/2030	615,000.00	5.000%	112.320	690,768.00
SERIAL	12/01/2031	750,000.00	5.000%	112.216	841,620.00
SERIAL	12/01/2032	795,000.00	5.000%	111.904	889,636.80
SERIAL	12/01/2033	850,000.00	5.000%	111.645	948,982.50
SERIAL	12/01/2034	950,000.00	5.000%	111.283	1,057,188.50
SERIAL	12/01/2035	1,000,000.00	5.000%	111.077	1,110,770.00
SERIAL	12/01/2036	1,075,000.00	5.000%	110.871	1,191,863.25
SERIAL	12/01/2037	1,175,000.00	5.000%	110.666	1,300,325.50
SERIAL	12/01/2038	1,250,000.00	5.000%	110.512	1,381,400.00
TERM45	12/01/2039	1,300,000.00	5.000%	109.444	1,422,772.00
TERM45	12/01/2040	1,450,000.00	5.000%	109.444	1,586,938.00
TERM45	12/01/2041	1,525,000.00	5.000%	109.444	1,669,021.00
TERM45	12/01/2042	1,600,000.00	5.000%	109.444	1,751,104.00
TERM45	12/01/2043	1,750,000.00	5.000%	109.444	1,915,270.00
TERM45	12/01/2044	1,850,000.00	5.000%	109.444	2,024,714.00
TERM45	12/01/2045	1,950,000.00	5.000%	109.444	2,134,158.00
TERM48	12/01/2046	2,100,000.00	5.000%	108.940	2,287,740.00
TERM48	12/01/2047	2,225,000.00	5.000%	108.940	2,423,915.00
TERM48	12/01/2048	2,350,000.00	5.000%	108.940	2,560,090.00
TERM54	12/01/2049	2,525,000.00	5.250%	109.580	2,766,895.00
TERM54	12/01/2050	2,675,000.00	5.250%	109.580	2,931,265.00
TERM54	12/01/2051	2,825,000.00	5.250%	109.580	3,095,635.00
TERM54	12/01/2052	2,975,000.00	5.250%	109.580	3,260,005.00
TERM54	12/01/2053	3,125,000.00	5.250%	109.580	3,424,375.00
TERM54	12/01/2054	3,275,000.00	5.250%	109.580	3,588,745.00
43,960,000.00					48,255,196.55

	Last Call Date	Issue Date	Remaining Weighted Average Maturity
School Facilities Improvement Bonds, Series 2018	12/01/2023	09/11/2018	26.1005
All Refunded Issues	12/01/2023		26.1005